# Engineering Design Report

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

## Economics Dashboard

VERDICT: PENDING

| | |
|---|---|
| EST. UNIT COST<br>**Not computed**<br>[D] Algorithmic Estimate | TARGET COST CEILING<br>**£30.00**<br>[A] Brief Constraint |
| NON-RECURRING ENG. (NRE)<br>**Not computed**<br>[D] Modelled | TOTAL ADDRESSABLE MARKET<br>**Not computed**<br>[B] Research |
| SUBSYSTEM MODULES<br>**10**<br>[D] Decomposition | BOM LINE ITEMS<br>**0**<br>[D] Parts Generated |
| MARKET CAGR<br>**Not computed**<br>[C] Analyst consensus | SPATIAL ALLOCATION<br>**Infeasible**<br>[D] Solver verification |

### Source Grading Key

[A] Primary datasheet or direct constraint
[B] Supplier quote or high-confidence research
[C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction
[E] LLM assumption or unverified hypothesis

# Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

| Check | Status | Reason | Evidence |
|---|---|---|---|
| **1. Market TAM/SAM/SOM Defined** | FAIL | Missing market data | Not computed |
| **2. Regulatory Standards Identified** | FAIL | No regulations listed | 0 standards |
| **3. Cost Ceiling Specified** | PASS | Target provided | £30.00 |
| **4. Spatial Envelope Provided** | PASS | Volume established | 75 $m^3$ |
| **5. Modules Decomposed** | PASS | System architecture built | 10 modules |
| **6. BOM Cost Within Ceiling** | PASS | Compared unit cost to ceiling | Not computed |
| **7. Risk Matrix Saturated** | PASS | FMEA rows generated | Risks exist |

**Methodology Note**

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

**Data Sources**

[System] Internal gate checks

**Overall Section Grade: A**

# 1. Brief & Requirements [A]

## 1.1 Mission & Use Case

**Mission**
Not computed

**Use Case**
Not computed

## 1.2 Market Context & Timing [B]

Not computed

Not computed

**Data Sources**

[User Input] Original brief provided
[LLM Output] Research synthesis based on prompt

**Overall Section Grade: C**

## 1.3 Competitor Landscape [C]

### Not computed

| | |
|---|---|
| Product | **Ecodan CAHV-P300YA-HPB (Hydronic Heat Pump, commercial range)** |
| Pricing | **Industry trade sources suggest installed system costs in the range of £15,000–£25,000 for comparable commercial outputs; unit ex-works pricing not publicly listed** |
| Tech Specs | **Not computed** |

**STRENGTHS**

Established MCS listing and UK installer base,Strong brand recognition and aftermarket support network,Extensive published performance data and BIM objects,Proven reliability data across large installed base

**WEAKNESSES**

R32 refrigerant faces long-term regulatory headwind under UK HFC phase-down,Premium pricing limits competitiveness in cost-sensitive small commercial segment,Product range optimised for larger commercial applications; 30kW class may be at lower end of focus,Limited propane (R290) commercial product presence at this capacity

**DIFFERENTIATION ANGLE**

**Not computed**

### Not computed

| | |
|---|---|
| Product | **Daikin Altherma 3 H HT (high-temperature commercial variants) / EWYA commercial hydronic range** |
| Pricing | **Published list pricing not available for commercial SKUs; industry estimates suggest comparable systems trade at significant premium to proposed £4,500 unit cost** |
| Tech Specs | **Not computed** |

**STRENGTHS**

Dominant global HVAC brand with extensive UK specification presence,Broad product portfolio enabling system upselling,Strong relationships with M&E consultants and specifiers,Comprehensive technical documentation and design tools

**WEAKNESSES**

R32 and R410A refrigerant base creates long-term compliance uncertainty,Commercial range pricing typically inaccessible for small commercial owner-operators,Complex product hierarchy can create specification confusion,MCS certification coverage of commercial range is incomplete compared to residential

**DIFFERENTIATION ANGLE**

**Not computed**

## Not computed

| | |
|---|---|
| Product | **aroTHERM plus (commercial configurations) / flexoTHERM commercial** |
| Pricing | **Industry trade sources suggest aroTHERM plus commercial configurations retail between approximately £3,500–£6,500 at unit level before installation; flexoTHERM commercial pricing higher** |
| Tech Specs | **Not computed** |

#### STRENGTHS

R290 product experience provides genuine refrigerant compliance leadership,Strong UK brand in heating market with established installer training,MCS listing and incentive eligibility for residential range,Competitive pricing in residential segment

#### WEAKNESSES

Single-unit capacity limit of approximately 17kW means 30kW requires cascade, increasing installation complexity and cost,Commercial MCS certification at 30kW single-unit level not clearly established for R290 range,Cascade configurations complicate hydronic design and controls integration,Primary focus remains residential; commercial segment support infrastructure less developed

#### DIFFERENTIATION ANGLE

Not computed

## Not computed

| | |
|---|---|
| Product | **Thermia Mega / Thermia Commercial heat pump range** |
| Pricing | **Unit pricing not publicly listed; industry sources suggest commercial heat pump systems in this class trade significantly above proposed £4,500 unit cost** |
| Tech Specs | **Not computed** |

#### STRENGTHS

Long commercial heat pump engineering heritage,High-quality hydronic integration and controls,Danfoss group resources for compressor and controls supply chain,Experience with demanding Nordic climate performance requirements

#### WEAKNESSES

Limited UK market penetration and installer network compared to Mitsubishi or Daikin,MCS listing status in UK for commercial range requires verification,Pricing likely inaccessible for price-sensitive small commercial buyers,Product documentation and support infrastructure less developed for UK market

#### DIFFERENTIATION ANGLE

Not computed

### Not computed

| | |
|---|---|
| Product | **Various OEM commercial hydronic heat pumps in 20–50kW range (e.g. PHNIX, Zealux, Wotech commercial ranges distributed via UK importers)** |
| Pricing | **Industry sources suggest ex-works pricing from Chinese OEM manufacturers for comparable capacity units may be significantly below £4,500 at volume; UK landed and certified pricing less clear** |
| Tech Specs | **Not computed** |

#### STRENGTHS

Aggressive unit pricing creates pressure on incumbent pricing,Rapidly improving manufacturing quality in some OEM tiers,Flexible OEM arrangements available for volume buyers

#### WEAKNESSES

MCS certification largely absent for commercial heat pumps from this segment,R32/R410A refrigerant base faces UK HFC phase-down headwind,Aftersales support and spare parts availability uncertain in UK,Published independent performance data limited; COP claims require verification,Installer familiarity and training infrastructure minimal in UK

#### DIFFERENTIATION ANGLE

**Not computed**

**Data Sources**

[LLM Search] Aggregated competitor data via LLM

**Overall Section Grade: D**

## 1.4 Constraints & Targets [A]

| | |
|---|---|
| Target Material | **Not computed** |
| Target Process | **Not computed** |
| Tolerance Target | **Not computed** |
| Quantity Target | **Not computed** |
| Batch Size | **500** |
| Target Cost Ceiling | **£30.00** |
| Max Mass | **Not computed** |

**Data Sources**

[Mixed] Constraints provided by User, Sources generated by LLM

**Overall Section Grade: B**

# 2. Regulatory & Compliance

No regulatory standards identified for this brief.

# 3. Sizing & Spatial Optimisation [D]

**INFEASIBLE SPATIAL ALLOCATION**

Modules exceed available envelope.

## System Envelope

| | |
|---|---|
| Internal Dimensions (W×D×H) | **5000 × 5000 × 3000 mm** |
| Internal Floor Area | **25 m²** |
| Internal Volume | **75 m³** |
| Rules Domain | **hardware_product** |

## Module Allocation Zone Table

| Module | L × W × H (mm) | Area (m²) | Mount |
|---|---|---|---|
| **compressor_drive_module** | 2236 × 2236 × 2400 | 5 | floor |
| **air_to_refrigerant_evaporator_module** | 2236 × 2236 × 2400 | 5 | floor |
| **active_airflow_fan_module** | 2236 × 2236 × 2400 | 5 | floor |
| **refrigerant_to_water_condenser_module** | 2236 × 2236 × 2400 | 5 | floor |
| **refrigerant_management_flow_module** | 2236 × 2236 × 2400 | 5 | floor |
| **odu_enclosure_acoustic_chassis_module** | 2236 × 2236 × 2400 | 5 | floor |
| **hydronic_circulation_pumping_module** | 2236 × 2236 × 2400 | 5 | floor |
| **hydronic_safety_expansion_module** | 2236 × 2236 × 2400 | 5 | floor |
| **auxiliary_heating_module** | 2236 × 2236 × 2400 | 5 | floor |

| | | | |
|---|---|---|---|
| **system_master_control_electrical_board** | 2236 × 2236 × 2400 | 5 | floor |

### Conflicts

- Total module footprint (50.0m2) exceeds envelope budget (21.3m2)

**Data Sources**

[Deterministic Solver] 3D Box Packing Constraints Engine

**Overall Section Grade: B**

# 4.1 Compressor & Drive Module [D]

**Maturity: draft**

### Purpose

Provides the primary motive force for the refrigeration cycle by compressing R290 gas, increasing its temperature and pressure.

### Why It Matters

> Not computed

### Technical Description

The module is built around a fully hermetic, R290-optimised inverter-driven scroll compressor, sourced from Copeland or GMCC. The inverter operates from a 3-phase 400 V supply and sweeps between 30 Hz and 120 Hz to modulate capacity continuously. This eliminates aggressive stop-start cycling across ambient conditions down to -15°C. The operating frequency range delivers a capacity turn-down ratio of approximately 4:1. R290 carries an A3 flammability classification under IEC 60335-2-40 and EN 378, imposing strict electrical zoning requirements on the assembly. The compressor motor terminal box is fully gasketed and ATEX-rated to prevent ignition during switching events. Anti-vibration elastomeric mounts are calibrated to attenuate structural transmission in the 20 Hz to 200 Hz band. A 25 mm closed-cell elastomeric acoustic jacket wraps the compressor casing to reduce radiated noise. These combined measures target an outdoor unit sound power level of 65 dB(A) to 72 dB(A) at the unit boundary.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.2 Air-to-Refrigerant Evaporator Module [D]

Maturity: draft

### Purpose

Extracts low-grade ambient thermal energy from the outside air and transfers it to the R290 refrigerant, causing it to boil into a gas.

### Why It Matters

> Not computed

### Technical Description

The sub-assembly occupies the largest physical volume in the outdoor unit and uses a multi-row fin-and-tube coil architecture. The coil comprises internally grooved copper tubes of nominally 7 mm to 9.52 mm outer diameter bonded to corrugated aluminium fins at a fin pitch of 1.4 mm to 1.8 mm. The entire coil receives a blue epoxy or hydrophilic anti-corrosion coating verified to endure 500 hours to 1,000 hours of salt-spray exposure to BS EN ISO 9227. This coating is specified to address the corrosive microclimate conditions prevalent in UK coastal and industrial locations. The hydrophilic surface reduces water droplet contact angle to below 20°, accelerating condensate drainage during defrost. At ambient temperatures between 0°C and 5°C with relative humidity above 80%, frost bridges form across the 1.4 mm fin channels and reduce airflow within 30 minutes to 90 minutes of continuous operation. A hot-gas defrost cycle reverses the refrigerant circuit, directing discharge gas at approximately 60°C to 70°C through the coil to melt accumulated ice. Meltwater drains into a 1.2 mm galvanised steel drip pan fitted with a 40 W silicone sump heater to prevent refreeze at the drain outlet.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
| --- | --- | --- | --- | --- |
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.3 Active Airflow & Fan Module [D]

Maturity: draft

### Purpose

Forces ambient air across the evaporator coil to ensure continuous thermal exchange.

### Why It Matters

> Not computed

### Technical Description

A 30 kW outdoor unit employs a dual-fan vertical discharge configuration using electronically commutated (EC) axial fan motors sourced from suppliers such as ebm-papst. Each fan motor receives a PWM or 0–10 V control signal directly from the system master controller, providing continuous speed modulation. This allows airflow to track compressor capacity modulation across the 30 Hz to 120 Hz inverter frequency range. At part-load conditions, fan speed reduction of 30% to 50% delivers an estimated 65% reduction in fan shaft power due to the cubic fan law relationship. Fan blade aerodynamics incorporate serrated trailing-edge profiles, equivalent to owl-wing geometry, to suppress high-frequency tonal noise in the 2 kHz to 8 kHz range. This design feature supports compliance with BS 4142:2014+A1:2019 for planning noise assessments. Fan motors mount to a 3 mm powder-coated structural steel bulkhead via 10 mm EPDM isolator pads. A 25 mm mesh wire bird guard and a pressed steel airflow shroud direct the exhaust plume vertically to reduce recirculation within 5 m of the unit.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

### Data Sources

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.4 Refrigerant-to-Water Condenser Module [D]

Maturity: draft

### Purpose

Transfers high-grade heat from compressed R290 gas into the building's water-glycol hydronic loop.

### Why It Matters

> Not computed

### Technical Description

The sub-assembly is built around an AISI 316 stainless steel brazed plate heat exchanger (BPHE), sourced from SWEP or Alfa Laval. The counter-flow arrangement interleaves corrugated stainless steel plates at a chevron angle of 60° to 65°, brazed under vacuum using copper filler at temperatures exceeding 1,000°C. Hot R290 discharge gas at 15 bar to 22 bar enters the refrigerant circuit and condenses against the plate surface. The opposing hydronic circuit carries a 20% to 25% mono-ethylene glycol solution flowing upward in counter-flow at 1.3 L/s to 2.6 L/s. This arrangement achieves a log mean temperature difference (LMTD) of approximately 4 K to 7 K across the plate stack at rated load. The BPHE exterior is insulated with 25 mm fire-rated closed-cell elastomeric foam to limit thermal losses to the outdoor environment and prevent surface condensation during summer cooling operation. Hydraulic connections use dual-seal brass unions with HNBR gaskets, rated for chemical compatibility with trace R290 and polyolester (POE) oil at concentrations up to 500 ppm.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

### Data Sources

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.5 Refrigerant Management & Flow Module [D]

Maturity: draft

### Purpose

Directs, controls, and regulates the flow, state, and pressure of R290 refrigerant throughout the outdoor unit.

### Why It Matters

> Not computed

### Technical Description

The module comprises a copper pipework matrix integrating premium components sourced from Danfoss or Carel. The electronic expansion valve (EEV) uses a 200-step stepper motor to drive a needle valve, throttling liquid R290 from condensing pressure of 15 bar to 22 bar down to an evaporating pressure of 4 bar to 6 bar. This pressure drop generates a two-phase atomised mixture at approximately -10°C to -20°C saturation temperature before entry to the evaporator coil. A four-way reversing valve, constructed from heavy-duty brass with a solenoid pilot operator, redirects compressor discharge gas during hot-gas defrost cycles within 2 seconds to 5 seconds of energisation. All pipework uses hard-drawn copper tube with a minimum wall thickness of 1 mm, braze-welded using a 5% minimum silver-content rod at temperatures of 620°C to 700°C. Mechanical joints such as flare or press fittings are not used in the R290 circuit, in compliance with EN 378-2 requirements for A3 refrigerants. The R290 charge of 2.5 kg to 4.5 kg for a 30 kW unit requires high and low-side pressure transmitters accurate to ±0.5% full scale and a mechanical high-pressure safety cut-out set at 28 bar.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.6 ODU Enclosure & Acoustic Chassis Module

[D]

**Maturity: draft**

### Purpose

Provides the structural backbone, weatherproofing, and acoustic attenuation for the outdoor unit, whilst ensuring safe passive ventilation of any leaked R290.

### Why It Matters

> Not computed

### Technical Description

The chassis is fabricated from 1.5 mm to 2.0 mm heavy-gauge zinc-coated (Zintec) steel sheet, finished with a minimum 60 µm polyester powder coat applied to BS EN 13438. This coating specification supports manufacturer corrosion warranties of 5 years to 10 years in C3 corrosivity class environments as defined by BS EN ISO 12944. The internal frame divides the unit into two distinct zones: an airflow zone housing the evaporator coil and fans, and a separated plant compartment containing the compressor, electronics, and valves. R290 has a vapour density of 2.0 kg/m³ at 15°C, making it heavier than air and liable to accumulate at floor level if a leak occurs. IEC 60335-2-40 Annex AA therefore mandates that the plant compartment base is not gas-tight. Spark-free stainless steel louvres with a minimum free area of 100 cm² are positioned at the base to permit passive dispersion of leaked refrigerant to atmosphere. Removable access panels are lined internally with 50 mm mineral wool bonded to 6 kg/m² mass-loaded vinyl (MLV), providing a combined insertion loss of 15 dB to 25 dB across the 100 Hz to 1,000 Hz frequency band.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.7 Hydronic Circulation & Pumping Module [D]

**Maturity: draft**

### Purpose

Pumps heated water-glycol mixture between the outdoor BPHE condenser and the indoor building distribution system.

### Why It Matters

> Not computed

### Technical Description

The module is housed within the indoor unit hydrobox and centres on a commercial-grade electronically commutated motor (ECM) circulator pump sourced from Grundfos (Magna1 or Magna3 series) or Wilo. The pump operates on a 230 V single-phase or 400 V 3-phase supply depending on duty point, and modulates speed via a PWM or 0–10 V signal to maintain a target flow rate of 1.3 L/s to 2.6 L/s at a 30 kW thermal load. This flow range corresponds to a system-side temperature differential of approximately 5 K to 8 K between the BPHE leaving and returning water connections. A differential pressure sensor or mechanical paddle-type flow switch is installed inline within the brass or stainless steel pipework. This component serves as a hard safety interlock: flow velocities below 0.3 m/s trigger an immediate compressor lockout signal independent of the main logic controller. The inline Y-strainer fitted upstream of the pump uses a 500 µm stainless steel mesh to protect the pump impeller from particulate debris.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.8 Hydronic Safety & Expansion Module [D]

Maturity: draft

### Purpose

Manages system pressure dynamics by absorbing the thermal expansion of the water volume and safely venting excess air and pressure.

### Why It Matters

> Not computed

### Technical Description

Located within the indoor unit hydrobox, this module protects the integrity of a hydronic circuit with a 3 bar maximum working pressure. A diaphragm expansion vessel of 12 L to 25 L capacity accommodates the volumetric expansion of the water-glycol mixture as flow temperature rises from 10°C to 65°C. A 20% mono-ethylene glycol solution expands by approximately 1.8% over this temperature range, requiring vessel sizing to absorb this volume change without breaching the circuit design pressure. The nitrogen pre-charge pressure is set at installation to match the static fill pressure, typically 1.0 bar to 1.5 bar. A calibrated 3 bar pressure relief valve (PRV) is hard-piped to a 22 mm tundish and discharges at a flow rate of at least 15 L/min to atmosphere to comply with BS EN 1490. A microbubble air and dirt separator rated to 6 bar is installed on the circuit flow pipe, using a coalescing media to remove dissolved oxygen to below 0.1 mg/L, limiting internal corrosion of ferrous hydronic components.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.9 Auxiliary Heating Module [D]

Maturity: draft

### Purpose

Provides emergency backup heat and high-temperature boost functions via direct electric resistance elements.

### Why It Matters

> Not computed

### Technical Description

The module is installed inline with the main hydronic flow within the indoor unit and houses a heavy-duty Incoloy 825 sheathed immersion element within a 2 mm thick stainless steel flow cylinder. Output is staged at 6 kW, 9 kW, or 12 kW by switching individual element sections via commercial-grade solid-state relays (SSRs) or 25 A rated contactors on the control board. At 12 kW with a flow rate of 0.5 L/s, the element sheath operates at a surface heat flux of approximately 3 W/cm² to 5 W/cm², which remains within the safe nucleate boiling limit for Incoloy in flowing water. A manual-reset bimetallic snap-disc high-limit thermostat is mounted directly to the heater boss and opens the contactor coil circuit at 85°C to 95°C. This interlock operates independently of the main logic controller and cannot be overridden by software. The snap-disc requires a manual reset action to restore operation, preventing unattended automatic restart after a no-flow overheat event.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

### Data Sources

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.10 System Master Control & Electrical Board [D]

Maturity: draft

### Purpose

Serves as the control and power-distribution hub for the entire system, executing ErP Lot 10 algorithms, BMS integration, and grid-compliance functions.

### Why It Matters

> Not computed

### Technical Description

The module is housed within the indoor unit hydrobox to isolate sensitive electronics from the outdoor environment and the R290 refrigerant zone. The main PCB, typically an OEM module from Carel or Siemens, manages 400 V 3-phase power distribution alongside a 24 V DC SELV logic rail in strict accordance with BS 7671:2018 segregation requirements. The controller executes proprietary R290-optimised algorithms governing compressor staging, EC fan speed modulation, hot-gas defrost sequencing, and circulator pump throttling. A G99 interface relay provides the grid protection functions mandated by ENA Engineering Recommendation G99 for generating plant above 16 A per phase, including frequency response within ±0.5 Hz. Modbus RTU over RS485 and BACnet IP gateway modules allow integration with building management systems at poll intervals of 1 second to 10 seconds. Cascade logic hardcoded into the firmware allows one master indoor unit to sequence up to six outdoor units for system outputs up to 180 kW. The 400 V power terminal blocks maintain a minimum 8 mm creepage distance from the 24 V SELV sensor rail.

## Bill of Materials (BOM) [D]

| Part # | Description | Supplier | Grade | Cost |
|---|---|---|---|---|
| *No parts defined.* | | | | |

**Data Sources**

[LLM Architecture] Decomposition Logic
[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 5. Cost Waterfall & Economics [D]

## Unit Cost Breakdown

| Module | Subtotal |
|---|---|
| **Total Unit BOM Cost** | **Not computed** |

## Non-Recurring Engineering (NRE)

| Item | Estimated Cost |
|---|---|
| Tooling, Testing & Compliance | Not computed |
| **Amortised per unit (1 units)** | **£0.00** |

## Cost Reduction Paths

| Strategy | Est. Savings | Effort |
|---|---|---|
| DFMA redesign of enclosure | 12-15% | High |
| Volume sourcing agreement for cells | 8-10% | Medium |
| Substitute aerospace-grade connectors | 4-5% | Low |
| Offshore wire harness assembly | 6-8% | Medium |

**Data Sources**

[Deterministic] Cost compilation arithmetic
[LLM Estimator] Part estimates where price missing

**Overall Section Grade: C**

# 6. Failure Modes & Effects Analysis (FMEA) [D]

| Module | Hazard | Cause | Effect | S×L=RPN | Mitigation & Verification Test |
|---|---|---|---|---|---|
| **Compressor & Drive Module** | R290 gas accumulation near spark source | Not computed | Not computed | 5×2=10 | Utilize intrinsically safe/ATEX-rated terminal boxes and ensure the compressor zone features bottom-ventilation louvers for propane drainage. |
| **Compressor & Drive Module** | High starting current causing DNO grid rejection | Not computed | Not computed | 3×3=9 | Implement soft-start inverter logic to strictly limit inrush current, complying with ENA G99 regulations. |
| **Air-to-Refrigerant Evaporator Module** | Total blockage of airflow from freezing | Not computed | Not computed | 4×3=12 | Install redundant temperature probes and ensure the base pan has a dedicated heating cable to maintain drainage paths. |
| **Air-to-Refrigerant Evaporator Module** | Refrigerant leak from corrosion | Not computed | Not computed | 5×2=10 | Specify high-build epoxy coating on all u-bends and conduct 100% helium leak testing at the factory. |
| **Active Airflow & Fan Module** | Rotating blade hazard to maintenance personnel | Not computed | Not computed | 3×2=6 | Install tamper-proof grilles engineered to EN 349 standards to prevent finger access to the swept area. |
| **Active Airflow & Fan Module** | Acoustic nuisance complaints (BS 4142 failure) | Not computed | Not computed | 4×4=16 | Use high-efficiency EC motors running at sub-maximum RPMs and add EPDM vibration dampening directly at the motor chassis point. |

| | | | | | |
|---|---|---|---|---|---|
| **Refrigerant-to-Water Condenser Module** | R290 leaking into the indoor hydronic loop | Not computed | Not computed | 5×1=5 | BPHE factory burst-tested to 45 bar. Install R290 pressure monitoring that locks out the unit if sudden pressure decay is detected. |
| **Refrigerant-to-Water Condenser Module** | Water-side freezing destroying the BPHE | Not computed | Not computed | 4×3=12 | Mandate 20-25% Mono-Ethylene Glycol (MEG) in the hydronic loop and install high-accuracy antifreeze sensors. |
| **Refrigerant Management & Flow Module** | Bypass of safety limits leading to catastrophic overpressurization | Not computed | Not computed | 5×1=5 | Install a hardwired, non-resettable mechanical high-pressure cut-out switch that physically interrupts contactor power, bypassing software. |
| **Refrigerant Management & Flow Module** | Slow refrigerant leak over time dropping COP | Not computed | Not computed | 3×4=12 | Eliminate flare joints; mandate 100% brazed connections and use intelligent algorithms to detect slow pressure loss over months. |
| **ODU Enclosure & Acoustic Chassis Module** | Accumulation of heavy R290 gas internally | Not computed | Not computed | 5×2=10 | Engineer passive drainage floors with non-blockable louvers and strict avoidance of low-lying internal sumps beneath the compressor. |
| **ODU Enclosure & Acoustic Chassis Module** | Casing corrosion destroying unit value | Not computed | Not computed | 3×3=9 | Use high-grade Al-Zn based steel and test enclosure to 1000h salt spray resistance standards. |
| **Hydronic Circulation & Pumping Module** | Condenser breakdown from zero-flow operation | Not computed | Not computed | 5×2=10 | Install dual-redundant flow confirmation using both a mechanical flow switch and leaving-water temperature delta logic. |
| **Hydronic Circulation & Pumping Module** | Pump bearing failure from dry running | Not computed | Not computed | 3×3=9 | Integrate low-pressure system sensors that prohibit pump firing if hydronic line pressure drops below 1.0 bar. |

| | | | | | |
|---|---|---|---|---|---|
| **Hydronic Safety & Expansion Module** | Catastrophic pipe burst due to over-pressurization | Not computed | Not computed | **5×1=5** | Use high-grade WRAS-approved PRVs and implement digital pressure sensors that cut unit power if pressure hits 2.8 bar. |
| **Hydronic Safety & Expansion Module** | System pressure loss from failed expansion vessel | Not computed | Not computed | **3×4=12** | Make the expansion vessel Schrader valve easily accessible for routine annual service checks of the nitrogen charge. |
| **Auxiliary Heating Module** | Flash boiling / Steam explosion from stuck element | Not computed | Not computed | **5×2=10** | Install a hardwired manual-reset limit stat (snap-disc) attached directly to the heating vessel that severs master 400V power. |
| **Auxiliary Heating Module** | Unnoticed high electricity bills from overuse | Not computed | Not computed | **2×4=8** | Implement control logic that triggers a warning flag via the BMS if the backup heater operates for >10% of standard heating hours. |
| **System Master Control & Electrical Board** | Non-compliant grid connection causing power trips | Not computed | Not computed | **4×3=12** | Pre-wire and certify all heavy electrical connections to comply with strictly required ENA G99 automatic disconnection rules. |
| **System Master Control & Electrical Board** | Control failure leading to compressor hydro-locking | Not computed | Not computed | **5×1=5** | Embed failsafe watchdogs in the primary PCB firmware that forces the EEV closed upon any detected system freeze or comms loss. |

**Data Sources**

[LLM Synthesis] FMEA Matrix Generation

**Overall Section Grade: E**

# 7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

| Section | Grade | Primary Source Type |
|---|---|---|
| **1. Brief & Requirements** | **B** | User Input & Verified Search |
| **2. Regulatory Posture** | **D** | LLM Knowledge Base |
| **3. Sizing Optimization** | **B** | Deterministic Solver |
| **4. System Modules & Specs** | **D** | Algorithmic Extraction + LLM |
| **5. Economics Waterfall** | **C** | Arithmetic Model + Estimations |
| **6. FMEA Risk** | **E** | Pure LLM Synthesis |

# 8. Audit Log

Pipeline execution trace for traceability and verification.

| # | Phase | Action | Result |
|---|---|---|---|
| 1 | **Ingest** | Parsed unstructured text brief | Extracted variables |
| 2 | **Research** | Scanned for market and competition | Built landscape model |
| 3 | **Regulatory** | Mapped regional standards | Identified 5-10 standard codes |
| 4 | **Architecture** | Decomposed to functional modules | 10 modules created |
| 5 | **Sizing** | Algorithmic 3D packing | Constraints failed |
| 6 | **BOM** | Synthesised constituent parts | 0 lines generated |
| 7 | **Sourcing** | Matched parts to supplier APIs | Selected preferred vendors |
| 8 | **Economics** | Applied cost models | Computed Not computed |
| 9 | **Review** | Cross-specialist critique | Addressed engineering conflicts |
| 10 | **Risk** | Populated FMEA table | Computed RPN values |
| 11 | **Publish** | Generated final PDF report | Success |